The Fourier Transform, Typed

A matrix–vector product, a fast factorization, and a change of basis that changes what coordinates mean

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Where This Picks Up

Across the binding decks, the Fourier transform has done two jobs. It makes convolution cheap, and it explains why convolution commutes:

$$\mathcal{F}(\mathbf{u} \circledast \mathbf{v}) \;=\; \mathcal{F}\mathbf{u} \odot \mathcal{F}\mathbf{v}$$

So far it has been a tool. This deck gives it the treatment the fold just received: an explicit matrix–vector product, a type, and a reading of what its outputs mean.

Three questions for this deck

1. In what precise sense is the FFT a matrix–vector product – and why is it fast?
2. What type is $\mathcal{F}$, and what does it do to vectors, to matrices, and to the binding tensor?
3. When do Fourier coefficients *mean* something?

Fourier coefficients themselves were introduced in the bonus deck; the first slides here restate what is needed.

Roadmap

I. The DFT as a **matrix–vector product** – and the FFT as that same product, factored

II. $\mathcal{F}$ as a **typed function** – a change of basis acts on every index

III. **Compositional behavior and meaning** – what $\mathcal{F}$ preserves, what it translates, and when its coefficients carry meaning

Running examples: $\mathbf{u} = (1, 2, 0, 1)$ and $\mathbf{v} = (2, 0, 1, 1)$; Session 11b's model $\mathcal{D}_e = \{\text{Alice}, \text{Bob}, \text{Carol}, \text{Fido}\}$; and one new domain, the four seasons.

Part I
The Fourier Transform as a Matrix–Vector Product

The DFT Matrix, Row by Row

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

The discrete Fourier transform is a 4×4 matrix of powers of $-i$. Row k is wave k, conjugated, and each output is the inner product of $\mathbf{x}$ with one row:

$$\mathcal{F} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{pmatrix}$$

k	row k of $\mathcal{F}$, dotted with $\mathbf{u} = (1, 2, 0, 1)$	$(\mathcal{F}\mathbf{u})_k$
0	$(1, 1, 1, 1) \cdot \mathbf{u} = 1 + 2 + 0 + 1$	4
1	$(1, -i, -1, i) \cdot \mathbf{u} = 1 - 2i - 0 + i$	$1 - i$
2	$(1, -1, 1, -1) \cdot \mathbf{u} = 1 - 2 + 0 - 1$	-2
3	$(1, i, -1, -i) \cdot \mathbf{u} = 1 + 2i - 0 - i$	$1 + i$

A matrix–vector product in the plainest sense: four rows, four inner products, and in general n^2 multiplications. Wave k itself is $\mathbf{f}_k = (1,\ i^k,\ i^{2k},\ i^{3k})$ – it turns k quarter turns per step.

Nothing Is Lost: The Inverse Is the Conjugate

Two facts about the rows of $\mathcal{F}$. They are orthogonal, and each has squared length $n = 4$, using the complex inner product, which conjugates one side:

$$\mathcal{F}\,\overline{\mathcal{F}} = 4\,I \qquad \Longrightarrow \qquad \mathcal{F}^{-1} = \tfrac{1}{4}\,\overline{\mathcal{F}}$$

The inverse transform is the same matrix with i replaced by $-i$, divided by 4. And $\mathcal{F}$ is symmetric, so its columns are the same waves as its rows.

Compare the fold

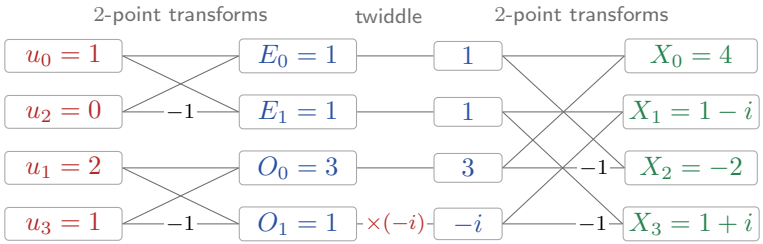
The fold's matrix Φ also had orthogonal rows of equal length: $\Phi\Phi^{\top} = 4I$. But Φ is 4×16, and twelve directions of its input fold to zero. $\mathcal{F}$ is 4×4, and none do.

Same orthogonality, opposite outcome: the fold compresses, while $\mathcal{F}$ only re-describes. Every vector can be recovered from its Fourier coefficients.

The FFT: The Same Product, Computed in Stages

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

Split $\mathbf{x}$ into its even and odd positions and give each half a 2-point transform, $(a, b) \mapsto (a + b,\ a - b)$. Multiply O_k by $(-i)^k$, then do 2-point transforms once more. Here is $\mathbf{u}$:



The evens $(1, 0)$ become $(E_0, E_1) = (1, 1)$ and the odds $(2, 1)$ become $(O_0, O_1) = (3, 1)$. The twiddle changes only O_1, to $-i$. Then $(E_0, O_0) \mapsto (X_0, X_2) = (4, -2)$ and $(E_1, -i) \mapsto (X_1, X_3) = (1-i,\ 1+i)$. Same four numbers as the dense product.

The FFT as a Chain of Sparse Matrices

Written out, the stages are three matrices, applied right to left:

$$\mathcal{F}_4 \;=\; B \cdot (\mathcal{F}_2 \oplus \mathcal{F}_2) \cdot P$$

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \qquad \mathcal{F}_2 \oplus \mathcal{F}_2 = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix} \qquad B = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -i \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & i \end{pmatrix}$$

P sorts the positions into evens and odds; the middle factor does the first 2-point transforms; B is the twiddle and the second 2-point transforms together. Each factor has at most two nonzeros per row. The FFT is matrix–vector multiplication, carried out through a chain of sparse matrices.

The order of the chain matters. These are functions being composed, and $P \cdot (\mathcal{F}_2 \oplus \mathcal{F}_2) \cdot B$ is not $\mathcal{F}_4$ – the lesson Session 11b drew from stacking alleged and former.

Why the Chain Is Short

Each stage halves the problem: a transform of size n is two of size $n/2$ plus $n/2$ combining steps. After $\log_2 n$ stages the cost is about $(n/2)\log_2 n$ multiplications, instead of n^2:

n	dense product, n^2	FFT, $(n/2)\log_2 n$
4	16	4
512	262,144	2,304
1,024	1,048,576	5,120
2,048	4,194,304	11,264

The algorithm is Cooley and Tukey's (1965). It is what makes HRR practical at high dimension: a binding costs two forward transforms, n products, and one transform back.

At $n = 4$ every multiplication is by ± 1 or $\pm i$, so the counts are only bookkeeping; the savings matter as n grows.

Part II
$\mathcal{F}$ as a Typed Function

What Type Is $\mathcal{F}$?

$\mathcal{F}$ takes a vector to a vector: type $\langle e, e\rangle$, order 2 – a *function*, like quickly, with rows for outputs and columns for inputs. But it is a special kind of function: invertible, and best read *passively*.

Actively, $\mathcal{F}$ sends $\mathbf{x}$ to a new vector $\mathcal{F}\mathbf{x}$. **Passively**, $\mathcal{F}\mathbf{x}$ lists the coordinates of the *same* $\mathbf{x}$ with respect to the waves instead of the individuals.

function	**type**	**invertible?**	**an individual goes to**
quickly, Q	$\langle\langle e, t\rangle, \langle e, t\rangle\rangle$	not in general	– (it acts on predicates)
relabeling P: Alice $\leftrightarrow$ Bob	$\langle e, e\rangle$	yes	another individual
$\mathcal{F}$	$\langle e, e\rangle$	yes	a wave across all four coordinates

Relabeling and $\mathcal{F}$ are both **changes of basis**. Relabeling keeps one coordinate per individual; $\mathcal{F}$ does not. That difference drives all of Part III.

Strictly, $\mathcal{F}$ maps $\mathbb{R}^4$ into $\mathbb{C}^4$: complex numbers are the price of a basis made of waves. A real vector always lands where coefficient $-k$ is the conjugate of coefficient k.

A Change of Basis Acts on Every Index

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

When the coordinates change, every index of every object changes with them. A vector has one index and needs one copy of $\mathcal{F}$; a function of type $\langle e, e \rangle$ has an output index and an input index, and needs two:

$$\mathbf{x} \;\mapsto\; \mathcal{F}\mathbf{x}, \qquad\qquad A \;\mapsto\; \mathcal{F}\,A\,\mathcal{F}^{-1}$$

Worked, at order 2. The circulant $C(\mathbf{u})$ – binding with $\mathbf{u}$ – becomes diagonal:

$$\mathcal{F}\,C(\mathbf{u})\,\mathcal{F}^{-1} \;=\; \mathrm{diag}(\mathcal{F}\mathbf{u}) \;=\; \mathrm{diag}(4,\; 1{-}i,\; -2,\; 1{+}i)$$

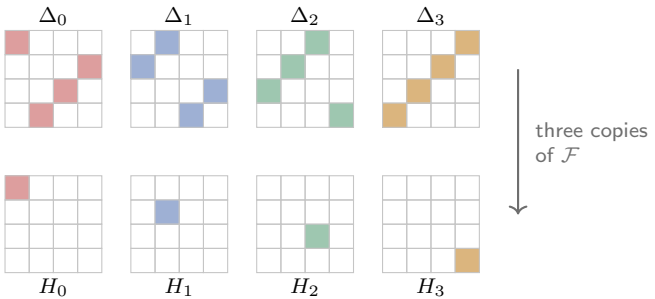
In the Fourier basis, binding with $\mathbf{u}$ multiplies coefficient k by $(\mathcal{F}\mathbf{u})_k$, one coordinate at a time.

So Session 11b's rule, *order = number of indices*, gains a second meaning: the order is the number of copies of the change of basis that an object needs.

At Order 3: The Binding Tensor Becomes the Hadamard Tensor

Δ has three indices – one output, two inputs – so it needs three copies:

$$\sum_{a,b,c} \mathcal{F}_{ka}\, \Delta_{abc}\, (\mathcal{F}^{-1})_{bi}\, (\mathcal{F}^{-1})_{cj} \;=\; H_{kij} \;=\; [\,i=j=k\,]$$



The four permutation masks become four single cells on the diagonal. So $\mathcal{F}(\mathbf{u} \circledast \mathbf{v}) = \mathcal{F}\mathbf{u} \odot \mathcal{F}\mathbf{v}$ is not a trick: it is the change-of-basis law for an order-3 object. Convolution and the Hadamard product are one tensor, seen in two coordinate systems.

Same Order, Different Laws: The Identity Matrix Three Ways

Session 11b warned that order counts indices but not what they mean: likes and quickly are both $n \times n$. A change of basis can tell them apart, because it treats arguments and results differently. Take the identity matrix I:

I **read as**	**type**	**law**	**in the Fourier basis**
a function: "do nothing"	$\langle e, e \rangle$	$\mathcal{F}\, I\, \mathcal{F}^{-1}$	I
a bilinear form: the dot product	$\langle e, \langle e, t \rangle\rangle$, like likes	$\mathcal{F}^{-\top}\, I\, \mathcal{F}^{-1}$	$\frac{1}{4}R$
a superposition of pairs, $\sum_i \mathbf{e}_i \otimes \mathbf{e}_i$	$e \otimes e$	$\mathcal{F}\, I\, \mathcal{F}^{\top}$	$4R$

Here R reverses a vector, $(R\mathbf{x})_k = x_{-k}$ – it is the fold's mask Δ_0. One matrix, three types, three answers.

"Doing nothing" looks the same in every basis. The dot product does not: in Fourier coordinates it pairs frequency k with frequency $-k$, $\mathbf{x} \cdot \mathbf{y} = \frac{1}{4} \sum_k (\mathcal{F}\mathbf{x})_k (\mathcal{F}\mathbf{y})_{-k}$. The law sees what order alone cannot – which indices are arguments and which are results.

Individuals Are Unitary

Transform a one-hot individual: $\mathcal{F}\mathbf{e}_a = \big(1,\ (-i)^a,\ (-1)^a,\ i^a\big)$ – every coefficient has magnitude 1. In Plate's terms every individual is a **unitary** vector, so its involution is its exact inverse: $\mathbf{e}_a^* \circledast \mathbf{e}_a = \boldsymbol{\delta}$. In the indicator regime, unbinding is exact; the trouble there was collisions, not noise.

More generally, no vector is concentrated in both bases. Donoho and Stark: $|\operatorname{supp}\mathbf{x}| \cdot |\operatorname{supp}\mathcal{F}\mathbf{x}| \ \geq\ n$.

$\mathbf{x}$	nonzeros of $\mathbf{x}$	nonzeros of $\mathcal{F}\mathbf{x}$	product
an individual, $\mathbf{e}_2$	1	4	4
animal $= (1,1,1,1)$	4	1	4
$(1,0,1,0)$	2	2	4
$\mathbf{u} = (1,2,0,1)$	3	4	12

Localist in one basis means distributed in the other. The universal predicate animal becomes "one-hot" at frequency 0: $\mathcal{F}(1,1,1,1) = (4,0,0,0)$.

A Predicate, Two Ways

The vector $(1, 1, 1, 0)$ for student has two readings, and the change of basis tells them apart:

read as	**type**	**law**	**in the Fourier basis**
the bundle Alice + Bob + Carol	e	$\mathcal{F}\mathbf{p}$	$(3,\ -i,\ 1,\ i)$
the function true of the students	$\langle e, t\rangle$	$\mathcal{F}^{-\top}\mathbf{p}$	$\frac{1}{4}(3,\ i,\ 1,\ -i)$

The function's law is the one that keeps application working. Contract its new coordinates with $\mathcal{F}(\text{Alice}) = (1, 1, 1, 1)$: $\frac{1}{4}(3 + i + 1 - i) = 1$, true. With $\mathcal{F}(\text{Fido}) = (1,\ i,\ -1,\ -i)$: $\frac{1}{4}(3 - 1 - 1 - 1) = 0$, false. Application is still contraction in the new basis, provided each object transforms by the law of its type. The bundle's coordinates, contracted the same way, give 4 for both.

The two rows are complex conjugates, up to the factor n. That is why the complex inner product conjugates one side: conjugating turns a bundle's coordinates into a function's. Part III uses the bundle reading – the Fourier coefficients of a set are those of the sum of its members.

Part III
Compositional Behavior, and Meaning

What $\mathcal{F}$ Preserves: All the Geometry

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

Inner products survive, up to the factor n. This is **Parseval's identity**, with the complex inner product $\langle \mathbf{X}, \mathbf{Y} \rangle = \sum_k X_k \overline{Y_k}$:

$$\langle \mathcal{F}\mathbf{x}, \mathcal{F}\mathbf{y} \rangle \;=\; n \, \langle \mathbf{x}, \mathbf{y} \rangle$$

	original	**after $\mathcal{F}$**
squared length of $\mathbf{u}$	$1+4+0+1=6$	$16+2+4+2=24=4\cdot 6$
inner product of $\mathbf{u}$ and $\mathbf{v}$	$2+0+0+1=3$	$12=4\cdot 3$

Lengths scale by $\sqrt{n}$, and every cosine is unchanged. Every similarity judgment in this course survives the change of basis: cosine does not care which basis it is computed in.

Divide $\mathcal{F}$ by $\sqrt{n}$ and it preserves lengths exactly – a rotation of complex space.

A Dictionary: What $\mathcal{F}$ Turns Into What

in the position basis	**in the Fourier basis**
binding, $\mathbf{u} \circledast \mathbf{v}$	the coordinate-wise product $\mathcal{F}\mathbf{u} \odot \mathcal{F}\mathbf{v}$
correlation, $\mathbf{u} \star \mathbf{v}$	$\overline{\mathcal{F}\mathbf{u}} \odot \mathcal{F}\mathbf{v}$
the involution, $\mathbf{u}^*$	the complex conjugate, $\overline{\mathcal{F}\mathbf{u}}$
rotation by one place	multiplication by $(1,\ -i,\ -1,\ i)$
a circulant, $C(\mathbf{u})$	a diagonal matrix, $\mathrm{diag}(\mathcal{F}\mathbf{u})$
the binding tensor, Δ	the Hadamard tensor, H
the exact inverse, $\mathbf{u}^\dagger$	the reciprocal, $1 \oslash \mathcal{F}\mathbf{u}$
the sum of the entries	coefficient 0

Every binding operation becomes coordinate-wise – and bundling, $+$, was coordinate-wise already. That is the whole reason HRR is computed in the Fourier basis.

What $\mathcal{F}$ Does Not Preserve: What a Coordinate Means

In the indicator regime, coordinate i of a predicate answers one question: is individual i in the set? After $\mathcal{F}$, no coordinate is about any one individual; each mixes all four.

$$\mathcal{F}(\text{student}) = \mathcal{F}(1,1,1,0) = (3,\ -i,\ 1,\ i), \qquad \mathcal{F}(\text{sleeps}) = \mathcal{F}(1,0,1,1) = (3,\ i,\ 1,\ -i)$$

With Alice at 0, sleeps happens to be the involution of student, so the dictionary predicts complex conjugates – and they are.

$\mathcal{F}$ carries the indicator regime into coordinates that behave like the distributional regime's: "meaning nothing in particular," in Session 11b's phrase.

Exactly one coordinate is an exception, and it is the one the course has been using since Session 6.

Coefficient 0 Is a Quantifier

$(\mathcal{F}\mathbf{a})_0 = \sum_x a_x = |A|$ – the count every determiner in Session 6 used, $|A| = \mathbf{1}^\top \mathbf{a}$. For student and for sleeps it is 3. (Read as a function, coefficient 0 is $|A|/4$: the proportion of the domain in A.)

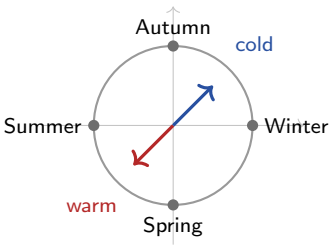
And it is the only coefficient that passes Tarski's test – for determiners, Session 6's ISOM: permute the domain, and nothing may change. Swap Alice and Bob in sleeps:

	before	**after the swap**
coefficient 0: how many sleepers	3	3
coefficient 2: sleepers at even positions minus sleepers at odd positions	1	-1

The other coefficients are arithmetic on names. The intersection count is still recoverable – $|A \cap B| = \frac{1}{4}\langle \mathcal{F}\mathbf{a}, \mathcal{F}\mathbf{b} \rangle = 2$ for student and sleeps – but it is spread across every frequency rather than stored in one.

When the Domain Has a Cycle: The Seasons

Number the seasons Winter 0, Spring 1, Summer 2, Autumn 3. Now "$+1$" means "one season later": the arithmetic belongs to the domain, not to the labels.



warm $= \{\text{Spring}, \text{Summer}\}$: $\mathcal{F}(\text{warm}) = (2,\ -1{-}i,\ 0,\ -1{+}i)$

cold $= \{\text{Autumn}, \text{Winter}\}$: $\mathcal{F}(\text{cold}) = (2,\ 1{+}i,\ 0,\ 1{-}i)$

Coefficient 0: how many seasons.

Coefficient 1: place season j at the point $(-i)^j$ on the circle; coefficient 1 is the sum of the members' points – $|A|$ times their center of mass, drawn here. It points at the predicate's peak: warm toward late spring, cold toward late autumn, in opposite directions.

Coefficient 3 is the conjugate of coefficient 1, frequency -1, as for every real vector.

A Season Later Is a Quarter Turn

Shifting a predicate one season later multiplies coefficient k by $(-i)^k$:

predicate	**Fourier coefficients**	**coefficient** 1
warm $=$ {Spring, Summer}	$(2,\ -1{-}i,\ 0,\ -1{+}i)$	–
one season later: {Summer, Autumn}	$(2,\ -1{+}i,\ 0,\ -1{-}i)$	a quarter turn
two seasons later: {Autumn, Winter} $=$ cold	$(2,\ 1{+}i,\ 0,\ 1{-}i)$	a half turn

The magnitudes never change: $(2,\ \sqrt{2},\ 0,\ \sqrt{2})$ in every row. A time shift moves the phases and nothing else.

"One season later" is a function of type $\langle e, e\rangle$ – a circulant, binding with $\mathbf{e}_1$ – and in the Fourier basis it is diagonal, $\mathrm{diag}(1, -i, -1, i)$.

Here the arithmetic of binding is the domain's own. Read $\mathbf{e}_j$ as "j seasons after Winter": then $\mathbf{e}_1 \circledast$ Spring $=$ Summer is a fact about the year. Bob $\circledast$ Carol $=$ Fido was a fact about nothing.

What Survives the Cycle's Symmetries

Which description of a seasonal predicate does not depend on which season we call 0, or on which way we go around the year? The magnitudes of its coefficients.

predicate	**pattern over the year**	**magnitudes** $\lvert\mathcal{F}\mathrm{x}\rvert$
warm $= \{\text{Spring}, \text{Summer}\}$	once a year	$(2,\ \sqrt{2},\ 0,\ \sqrt{2})$
solstice seasons $= \{\text{Winter}, \text{Summer}\}$	twice a year	$(2,\ 0,\ 2,\ 0)$

Rotating the year, or running it backward, leaves both rows unchanged. A relabeling that ignores the cycle – swapping Winter and Spring – turns warm into $\{\text{Winter}, \text{Summer}\}$, and its magnitudes into $(2, 0, 2, 0)$. Even the magnitudes mean only as much as the cycle does.

Tarski took his test from Klein, who classified geometric notions by the transformations they survive. Logic asks for survival under every relabeling; a domain with structure asks only for its own symmetries, and so keeps more. Individuals have no cycle, and only coefficient 0 survives; the seasons have one, and every magnitude survives.

Where the Cycle Is Real in Language: Word Positions

Positions in a sentence have the additive structure that individuals lack: three words later is three words later wherever you start. Transformers use exactly this.

Sinusoidal position encodings (Vaswani et al. 2017)

Position pos gets $\sin(pos/10000^{2i/d})$ and $\cos(pos/10000^{2i/d})$ – sinusoids at many frequencies. The authors expected this to let a model attend by relative position, since "for any fixed offset k, PE_{pos+k} can be represented as a linear function of PE_{pos}."

Rotary position embeddings (Su et al. 2021)

Each pair of query and key coordinates is rotated by an angle proportional to the token's position. The score between positions m and n then depends on position only through $e^{i(m-n)\theta}$ – only on the *relative* position.

Both are the seasons slide at scale: a shift is a phase rotation, so anything built from magnitudes or phase differences does not care where counting started.

Summary: The Fourier Transform, Typed

object	type, order	how $\mathcal{F}$ acts	in the Fourier basis
individual, role; a set as a bundle	e, 1	$\mathcal{F}\mathbf{x}$	individuals become waves of magnitude 1
predicate, as a function	$\langle e, t\rangle$, 1	$\mathcal{F}^{-\top}\mathbf{p}$	the bundle's coefficients, conjugated, over n
function: $C(\mathbf{u})$, a time shift	$\langle e, e\rangle$, 2	$\mathcal{F}A\mathcal{F}^{-1}$	circulants become diagonal
relation, bilinear form	$\langle e, \langle e, t\rangle\rangle$, 2	$\mathcal{F}^{-\top}G\mathcal{F}^{-1}$	the dot product pairs k with $-k$
pair	$e \otimes e$, 2	$\mathcal{F}M\mathcal{F}^{\top}$	the fold becomes "keep the diagonal"
binding, Δ	$\langle e, \langle e, e\rangle\rangle$, 3	one $\mathcal{F}$, two $\mathcal{F}^{-1}$	the Hadamard tensor H

- The DFT is a matrix–vector product; the FFT computes it through a chain of $\log_2 n$ sparse factors.
- $\mathcal{F}$ is a change of basis. It preserves the geometry and makes every binding operation coordinate-wise.
- Order counts the copies of $\mathcal{F}$ an object needs, and the law tells arguments from results – what order alone could not.
- Coefficient 0 is a quantifier. The other coefficients mean something only when the domain has a cycle.

In-Class Exercises

1. Compute $\mathcal{F}(\text{student}) = \mathcal{F}(1,1,1,0)$ row by row, and check coefficient 0 against $|\text{student}|$.
2. Run the FFT by hand on $\mathbf{v} = (2,0,1,1)$: the even half, the odd half, the twiddle, then combine. Check your answer against $\mathcal{F}\mathbf{v} = (4,\ 1{+}i,\ 2,\ 1{-}i)$.
3. Compute the coefficients of the solstice seasons, $(1,0,1,0)$. What do their magnitudes say about the predicate's pattern over the year?
4. Shift warm by two seasons. What happens to each coefficient, and which predicate do you get?
5. Show that $\mathcal{F} I \mathcal{F}^{-1} = I$, and explain why the dot product, by contrast, changes form in the Fourier basis.

⋆6. Show that every circulant commutes with the one-place shift S. Then, using the fact that the waves are eigenvectors of S with four different eigenvalues, conclude that every circulant is diagonal in the Fourier basis.

In-Class Exercises: Answers

1. $(3,\ -i,\ 1,\ i)$. Coefficient 0 is $1+1+1+0=3=|\text{student}|$.

2. Evens $(2,1)\mapsto(3,1)$; odds $(0,1)\mapsto(1,-1)$. The twiddle turns $O_1=-1$ into $(-i)(-1)=i$. Then $(3,1)\mapsto(X_0,X_2)=(4,2)$ and $(1,i)\mapsto(X_1,X_3)=(1{+}i,\ 1{-}i)$.

3. $(2,\ 0,\ 2,\ 0)$: all the weight sits at frequencies 0 and 2, a pattern that repeats twice a year. warm's $(2,\sqrt{2},0,\sqrt{2})$ puts its weight at frequencies ±1: once a year.

4. Coefficient k is multiplied by $(-i)^{2k}=(-1)^k$: coefficients 1 and 3 change sign, while 0 and 2 stay. You get $(2,\ 1{+}i,\ 0,\ 1{-}i)=\mathcal{F}(\text{cold})$ – half a year after the warm seasons come the cold ones.

5. $\mathcal{F}I\mathcal{F}^{-1}=\mathcal{F}\mathcal{F}^{-1}=I$: a function that does nothing does nothing in every basis. The dot product is a bilinear form, which transforms by $\mathcal{F}^{-\top}I\mathcal{F}^{-1}=\frac{1}{4}R$, so in Fourier coordinates it pairs frequency k with $-k$.

6. $C(\mathbf{u})=\sum_s u_s S^s$, a sum of powers of S, so it commutes with S. If $S\mathbf{f}=\lambda\mathbf{f}$, then $S(C\mathbf{f})=C(S\mathbf{f})=\lambda\, C\mathbf{f}$; since each eigenvalue belongs to a single wave, $C\mathbf{f}$ is a multiple of $\mathbf{f}$. Every wave is an eigenvector of C, so C is diagonal in the wave basis.

Reading

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- Donoho & Stark, *Uncertainty principles and signal recovery*, SIAM Journal on Applied Mathematics 49, 1989.
- Plate, *Holographic reduced representations*, IEEE Transactions on Neural Networks, 1995 – unitary vectors.
- Vaswani et al., *Attention is all you need*, NIPS 2017 – sinusoidal position encodings.
- Su, Lu, Pan, Murtadha, Wen & Liu, *RoFormer: enhanced transformer with rotary position embedding*, arXiv 2021.
- Tarski, *What are logical notions?*, History and Philosophy of Logic 7, 1986.